A Theory of Privacy

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Overview

1. Why do we need (a theory of) privacy?
2. A microeconomic model
3. Four Results
4. Examples

What is Privacy?

- The ability to take actions without being observed, and having interactions with others confined to the intended recipients
- (There are many other definitions.)
- Privacy $\subset$ asymmetric information

Privacy: A hot topic

- Several Western governments systematically collect internet traffic data, metadata, read emails, can access webcams, ...
- People voluntarily provide lots of information on Facebook, Twitter, ...
- Giving up information for better prices (loyalty programs, insurance contracts)
- Modern administrations collect lots of data and make it accessible to many (register data!)
- $\Rightarrow$ We need to understand what's at stake, which effects privacy has, which trade-offs come with it

Economics of Privacy: The Chicago School

- The most well-known economic theory of privacy
- Exemplified by George Stigler (1980, JLS) and Richard Posner (1981, AER)
- Argument:
  - Privacy = asymmetric information
  - Asymmetric information leads to inefficiencies
  - Therefore: privacy = inefficient

Popular debate: The "Nothing to hide" argument

- "The most common retort against privacy advocates" (Schneier 2006)
- Used by people at NSA, Google, ...
- Argument:
  - If you have nothing to hide, you have nothing to fear from giving up your privacy
  - If you have something to hide, you should not be able to hide it!
- (Note the similarities to the Chicago argument.)

What we do

- We develop an informational theory of privacy: why do people care about it, what are the welfare effects?
- We do not consider a "taste for privacy" – instead, we derive the individual and social benefit of privacy theoretically
- The Chicago argument ignores two things:
  - Information is often not perfect and unambiguous (decisions are made by statistical discrimination; Phelps 1972, Arrow 1973)
  - People change their behavior when they know they are observed
- It also compares with an imaginary first best (in reality, we can never live without asymmetric information $\Rightarrow$ privacy is about how it should be structured)

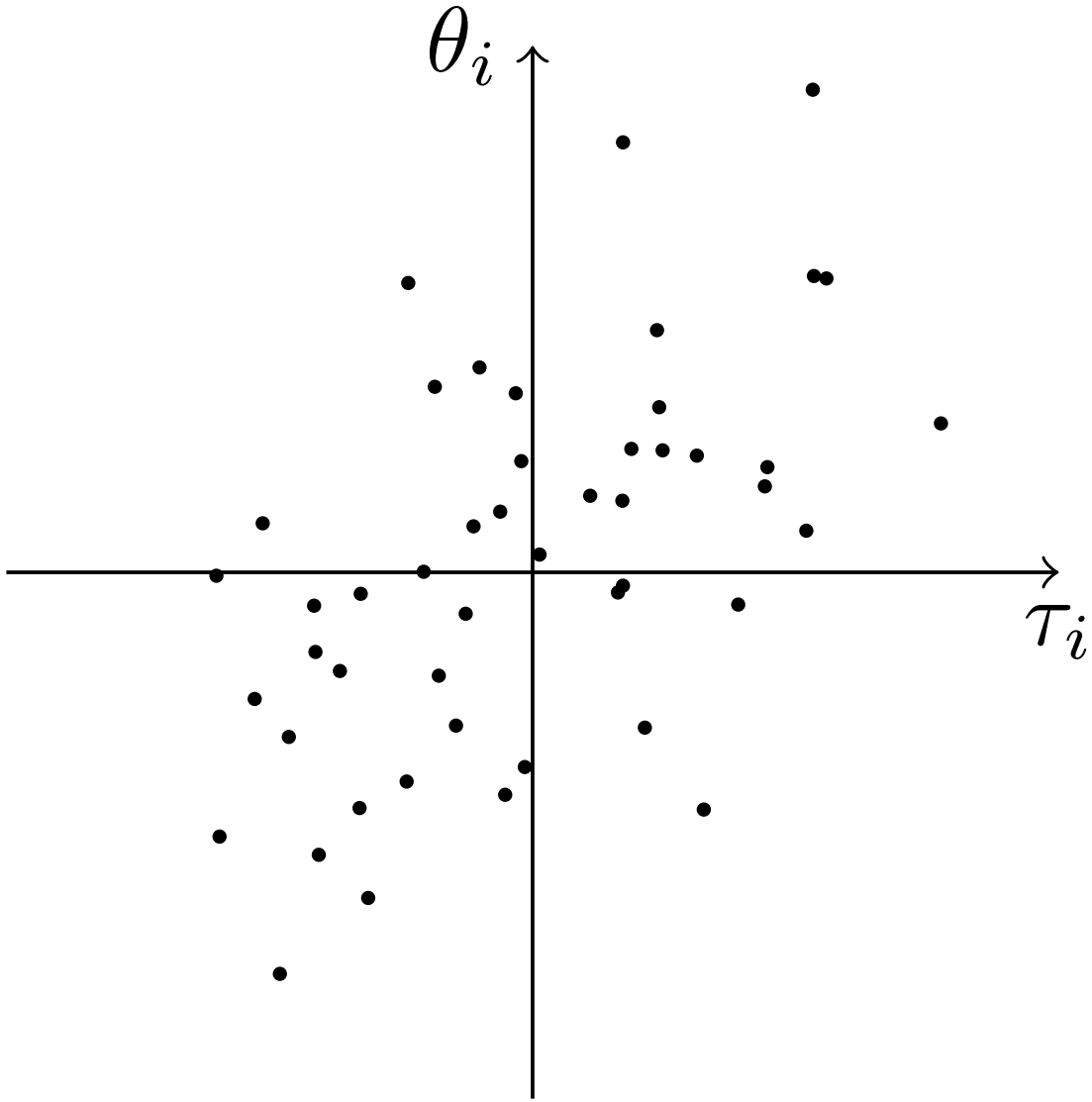
4 main results + 2 extensions

- Lack of privacy changes behavior systematically ("chilling effect")
- The chilling effect reduces overall welfare
- Effectiveness of privacy intrusions is easily overstated
- Privacy is redistributive – it affects different people differently
- Extensions:
  - Voluntary privacy might not work
  - Lack of privacy can erode mutual trust ($\Rightarrow$ further welfare reduction)

The Model

- A group of n individuals have preferences over two policies, $p = 0, 1$
- Each individual i has two types: θ_i and τ_i (both are in $\mathbb{R}$)
- θ_i is the policy preference (e.g. on drug legalization); $\theta_i \sim \Phi(0, 1)$
- τ_i is the hidden type (e.g. drug use)
- τ_i is correlated with θ_i (formally: distribution of τ_i given $\theta_i^{'}$ F.O.S.D. the distribution of τ_i given $\theta_i^{''}$ if $\theta_i^{'} > \theta_i^{''}$)
- There is an opposing player (OP) who gains from treating people differently based on τ_i

Correlation between θ_i and τ_i



An example

- Alice thinks drugs should be legalized and wants to write on her Facebook profile about that
- She is also looking for a job
- Employers do not want to hire drug users – but that is not observable
- There is a correlation between stand on legalization (θ_i) and drug use (τ_i)
- Two cases: The employer can see what Alice does online, or not
- (Acquisti and Fong 2015: Employers discriminate based on social media profiles)

Timing

- Two stages:
  1. Information aggregation: Each individual observes θ_i and τ_i, chooses $p_i = 0$ or $p_i = 1$. Policy $p = 1$ is implemented with probability $q(m/n)$, where m/n is the proportion of individuals that choose $p_i = 1$
  2. Interaction: An opposing player (OP) chooses, for each individual, whether to treat them aggressively (A) or mildly (M)
  3. (Payoffs are realized:)
     - For individuals: $p\theta_i - \mathbb{1}_{s(p_i)=A}\delta(\tau_i)$, δ is an increasing function
     - For OP: $\sum_i \mathbb{1}_{s(p_i)=A}\tau_i$

What do we mean by correlation?

- One attribute is *predictive* of another
- The causality doesn't matter!
- "In the social sciences, everything is correlated" (Meehl 1990)
- Hamburger and Wallsten (2005, LA Times) about how the Republican party targets voters:

  "Bourbon drinkers were more likely to be Republican, while gin was a Democrat's drink. ... Democrats preferred Volvos; Ford and Chevy owners were most likely Republicans. People with call-waiting service on their telephones were predominantly Republican."

Correlation and Machine Learning

Hello [redacted],

We thought you might be interested in knowing that customers who bought "200g*0.01g Mini Digital Pocket Scale for Jewelry Kitchen Gram Oz Ct" also bought these items.

Customers Also Bought...

100 Plastic Resealable Grip Seal Bags...

Sold by: AGOODBUYFROMME

100 Grip Seal Bags 2.25 x 3 Inch...

Sold by: Express Goods UK

250 Raw Filter TIPS card booklets...

Sold by: Martins Deals

S AND S Â£20 Pound Note Design...

100 Grip Seal Bags 1.5" x 2.5"

JUICY Juicy Jays Kingsize...

S AND S 750 Roaches Roach Filter...

How we solve the game

- We consider two cases:
  - Privacy: OP can not observe p_i, has to treat everybody the same (we assume: mild, M)
  - No Privacy: OP can observe p_i before choosing how to treat a person
- We show that individuals use cutoff strategies ($p = 1$ if $\theta_i > t(\tau_i)$)
- We consider cutoffs and OP choices under the two cases

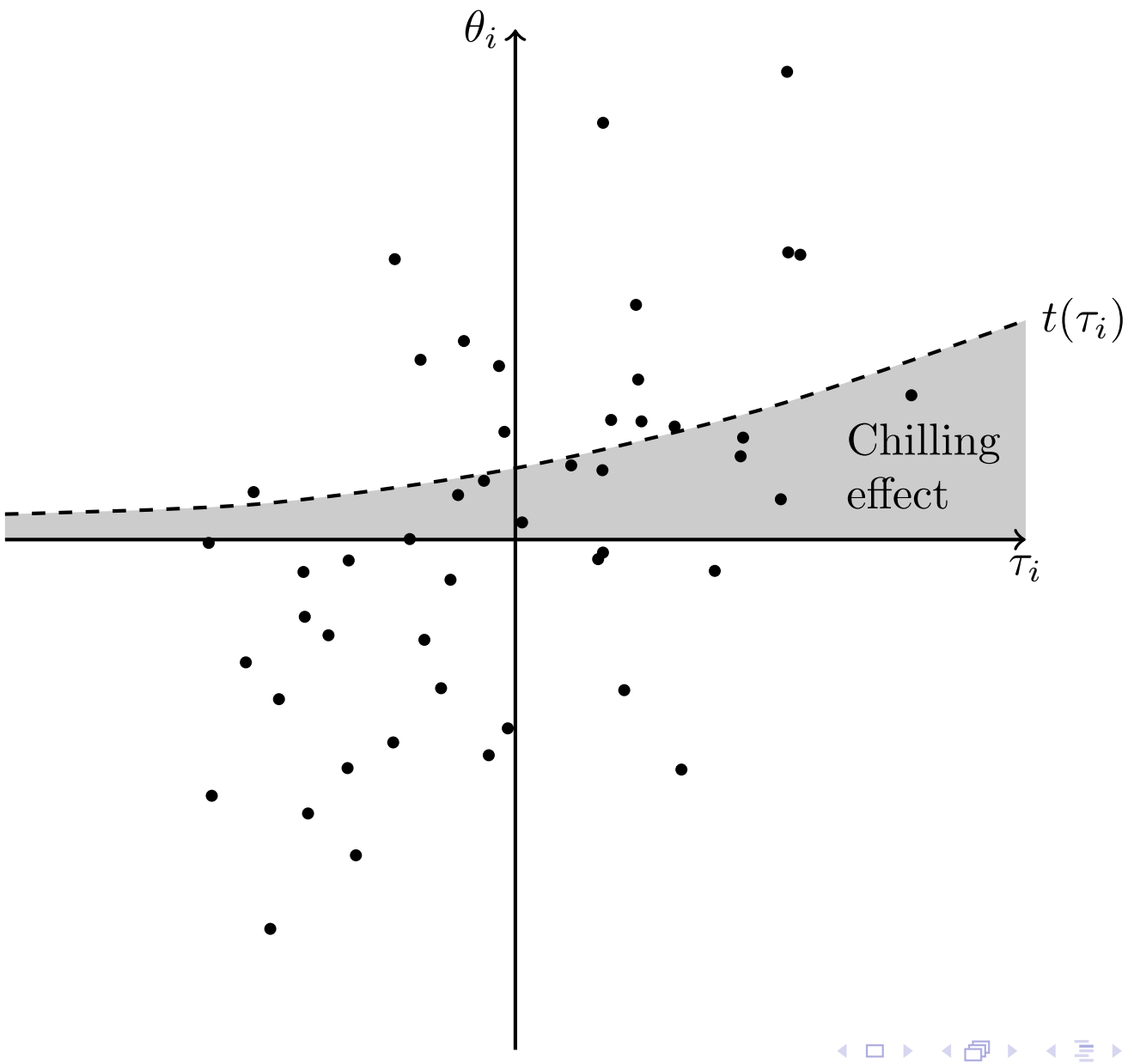
R1: The chilling effect

Proposition

For all i, the equilibrium cutoff $t(\tau_i)$ is higher without privacy. The difference $t^{\text{no privacy}}(\tau_i) - t^{\text{privacy}}(\tau_i)$ is increasing in τ_i.

- People with positive, but small θ_i choose $p_i = 0$ instead of 1
- OP still plays M against $p_i = 0$, but A with positive probability against $p_i = 1$

Chilling effect: Graph



R2: The chilling effect reduces welfare

Proposition

1) If OP uses a mixed strategy in the no-privacy equilibrium, then privacy welfare-dominates no privacy

2) If OP uses a pure strategy in the no-privacy equilibrium, then privacy welfare-dominates no privacy for large n. (+ technical assumptions)

- $\Rightarrow$ The chilling effect impairs information aggregation and reduces welfare
- (We consider welfare separately for the individuals and OP.)

Alice and the Chilling Effect

- Alice knows that she lowers her chances of being employed if she posts her opinion about drug legalization
- Therefore she doesn't (or misrepresents her views) $\Rightarrow$ chilling effect
- People who oppose legalization have no such problem
- The public debate is systematically skewed
- $\Rightarrow$ the welfare-optimal policy is less likely to be implemented, which lowers expected welfare

R3: Surveillance performs worse than expected*

- Naively, one could expect that taking away privacy (through surveillance, buying data, starting to use data that is already available) delivers large surplusses to the OP
- But the chilling effect reduces the correlation between p_i and τ_i
- Formally, we can show:

Proposition

OP's payoff without privacy is lower if citizens use the cutoffs $t^{\text{no privacy}}(\tau_i)$ than if they use $t^{\text{privacy}}(\tau_i)$.

R4: Privacy has different influences on different people

- Note that people with $\theta_i < 0$ are better off *without* privacy, since that makes it more likely that their preferred policy is implemented (and they always get treated mildly)
- People who are subject to the chilling effect are worse off
- People with $\theta_i > t^{\text{no privacy}}(\tau_i)$ are worst off: they lose $\delta(\tau_i)$ *and* their preferred option is less likely
- $\Rightarrow$ privacy is redistributive

Extension: Voluntary privacy

- Assume that individuals can choose whether OP can observe their p_i or not
- There are several equilibria:
  1. Everybody chooses privacy, OP plays M
  2. Only people with $p_i = 1$ choose privacy, or nobody chooses privacy $\Rightarrow$ as before
  3. (Some intermediary equilibria)
- 1 is not stable under the D1 criterium (Banks and Sobel 1987)
- $\Rightarrow$ Privacy might not work if it is voluntary, it has to be imposed

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- (Similar to Schelling's (1960) argument about the secret ballot.)

Extension: Defensive actions

- Assume that each person can take a costly defensive action to lower the disutility of being treated aggressively by OP at the same time as choosing p_i
- This action also lowers the utility of OP
- Without privacy, a person choosing $p_i = 1$ also chooses the defensive action
- In our example: Alice could hire a lawyer if she supports legalization
- Everybody is worse off
- $\Rightarrow$ Lack of privacy can lead to a "spiral of distrust", further decreases welfare

Credit scoring

- A bank has to decide to whom to lend; repayment probability (τ_i) is not directly observable
- Consider two preferences (two different θ_i) that are predictive of τ_i: Taste for education ($\Rightarrow$ education level) and taste in music
- Low education and a preference for rap music predict low repayment probability
- (There are "social scoring" companies who collect data like that, Facebook has a patent on a method)

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- Should the bank be allowed to use data on music taste? ("Equal Credit Opportunity Act" outlaws "redlining" in the US)

Work in committees

- A committee is debating two policies (e.g. raise interest rates or not)
- The debate and vote can either be in secret or in public
- There is a correlation between policy preference θ_i and competence τ_i
- Members worry about being perceived as incompetent; advocate less radical positions

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- The Fed is forced to publish minutes of FOMC meetings since 1993; studies show an increase in conformity and a decrease of disagreement with the chairman (Meade and Stasavage 2008)
- Thomas Hoenig, President of the Kansas City Fed: "The tape has had some chilling effect on our discussions. I see a lot more people reading their statements."

Conclusion

- Privacy is a thorny issue with many facets
- We propose a simple model to understand what privacy is, why it is important for welfare *from an informational perspective*
- Our model crystallizes some policy questions:
  - Which data should be available, usable about people?
  - Is it enough to give them the option of privacy?
  - How open should government be?
  - If privacy is redistributive – should that influence how much privacy people get?

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  - How open should government be?
  - If privacy is redistributive – should that influence how much privacy people get?
  - Note: freedom of expression is also redistributive. Friedman (1972) argues that it should still be universal